1. True or False? For a given data set, we can directly calculate the bias and variance of a regularized regression model to see whether or not the decrease in variance is enough to offset the increase in bias. Based on this, we can choose an optimal λ.

↳ we want to use AIC / BIC

2. True or False? Since ridge regression always returns the full model (with all p predictors), its test MSE will always be smaller than that of Lasso.

↳ depending on the size of the model

3. True or False? When comparing models of the same size, adjusted R^2 and BIC will lead us to pick the same 'best' model.

↳ the penalty term will be the same

4. To properly incorporate a categorical variable with k categories into our model, we learned that we need $k-1$ dummy variables that take on values 0 or 1. Suppose we accidentally mess up and include k dummy variables instead (one for each category). What will happen to our least square regression estimates ($\hat{\beta}$) in this situation?

↳ perfect multicollinearity

5. Use the following output to carry out (1) subset selection, (2) forward selection, and (3) backward selection. The entire data set has 10,000 observations. For each algorithm, report the 'best' model of each size. **State and compute the criteria** you will use to select your final model. Report the 'best' model of each size in the table below. Report your final model for each algorithm.

Model	$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	
$Y \sim X_1$	13650.32	1355.43
$Y \sim X_2$	13691.11	
$Y \sim X_3$	13755.97	
$Y \sim X_4$	13712.95	
$Y \sim X_1 + X_2$	13549.8	
$Y \sim X_1 + X_3$	13516.38	1314.60
$Y \sim X_1 + X_4$	13514.86	1314.12
$Y \sim X_2 + X_3$	13554.16	
$Y \sim X_2 + X_4$	13490.86	1306.4
$Y \sim X_3 + X_4$	13598.66	
$Y \sim X_1 + X_2 + X_3$	13411.79	1282.87
$Y \sim X_1 + X_2 + X_4$	13490.59	
$Y \sim X_1 + X_3 + X_4$	13476.56	1303.79
$Y \sim X_2 + X_3 + X_4$	13474.19	
$Y \sim X_1 + X_2 + X_3 + X_4$	13410.86	1284.57

You may want to use a table to organize your results:

	Subset	Forward	Backward
M_1	X_1	X_1	X_1
M_2	$X_2 + X_4$	$X_1 + X_4$	$X_1 + X_3$
M_3	$X_1 + X_2 + X_3$	$X_1 + X_3 + X_4$	$X_1 + X_2 + X_3$
M_4	$X_1 + X_2 + X_3 + X_4$	$X_1 + X_2 + X_3 + X_4$	$X_1 + X_2 + X_3 + X_4$
Final Model	$X_1 + X_2 + X_3$	$X_1 + X_3 + X_4$	$X_1 + X_2 + X_3$